

Kunwoo Lee

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Education

- Master of Science in Mechanical Engineering – Research**, GPA: 3.91/4.0 May 2026
Carnegie Mellon University, Pittsburgh, PA
- Bachelor of Science in Mechanical Engineering**, Summa Cum Laude, GPA: 3.86/4.0 May 2024
Illinois Institute of Technology, Chicago, IL
- Bachelor of Science in Mechanical Engineering**, Honor Student, GPA: 4.43/4.5 Feb 2017 – Dec 2021
Inha University, Incheon, South Korea

Master's Research Experience – Carnegie Mellon University, PA

Next-Generation Accessible Motion Tracking Systems Aug 2025 – Present

Graduate Research Assistant & MS Research – Musculoskeletal Biomechanics Lab (Prof. Eni Halilaj)

- Developing a generative AI framework that reconstructs high-fidelity full-body kinematics from sparse sensors (e.g., VR headset and controllers) with robust generalization to out-of-distribution single-legged movements.
- Building the extrinsic auto-calibration pipeline for *DeepGaitLab*, an open-source markerless MoCap system, enabling rapid multi-view setup and optimizing inverse dynamics modules to reach kinetic accuracy comparable to marker-based laboratories.

Calibration-Free Multimodal Physical Therapy (PT) Assessment May 2025 – Aug 2025

Graduate Research Assistant – Musculoskeletal Biomechanics Lab (Prof. Eni Halilaj)

- Co-developed a multimodal pipeline that autocalibrates two RGB cameras and aligns uncalibrated IMU signals to body segments, removing manual setup and enabling scalable PT assessment.
- Trained a random forest classifier on kinematic outputs from our auto-calibrated, multi-view and fusion system; models exceeded the clinician agreement (66.8%), demonstrating performance at the level of expert consensus.

Validating Computer Vision-based Human Kinetics Estimation Sep 2024 – Aug 2025

Graduate Research Assistant & MS Research – Musculoskeletal Biomechanics Lab (Prof. Eni Halilaj)

- Extended a muscle-driven simulation framework by adding modules for asymmetric foot-contact handling and kinematic stabilization, enabling accurate simulation of unconventional movements.
- Evaluated commercial and open-source softwares for human kinetics estimation against the gold-standard marker-based approach, enabling more informed decisions for the trade-off of accessibility and precision.
- Introduced a novel validation framework by simulating marker registration uncertainty to quantify variability in marker-based methods, unveiling that state-of-the-art multiview kinetics estimation can achieve comparable accuracy under specific conditions.

sEMG-Based Keyboard-Free Typing with Deep Learning Jan 2025 – May 2025

Course Project – Intro to Deep Learning (11-785), Carnegie Mellon University

- Developed a transformer-based model that maps wrist sEMG signals to keyboard input, improving prediction accuracy by 25% compared to leading Time-Depth Separable Network baseline.
- Conducted large-scale deep learning experiments across HPC resources (PSC Bridges-2) using Slurm, ensuring efficient, reproducible training with PyTorch Lightning and Hydra.

Undergraduate Research Experience

CNN-Based Estimation of Airfoil Kinematics Jan 2023 – May 2024

Undergraduate Research Assistant (Prof. Scott Dawson), Illinois Institute of Technology, IL

- Designed and trained a convolutional neural network (CNN) using TensorFlow to predict airfoil pitching kinematics from instantaneous downstream wake snapshots, offering faster prediction from smaller measurement fields, achieving over 90% accuracy.
- Generated high-fidelity data using direct numerical simulation (DNS) of both periodic and non-periodic airfoil motions, enabling robust training and evaluation of the CNN model across diverse flow regimes.

Equity Analysis of Tax Increment Financing (TIF) Jan 2024 – May 2024

Course Project – Interprofessional Projects (IPRO 497), Illinois Institute of Technology, IL

- Uncovered disparities in the economic impact of TIF, showing disproportionate benefits for White, higher-income wards through comparative analysis of TIF and non-TIF districts across racial and income groups, using clustering methods.

- Presented findings to peers and faculty, sparking discussions on the equity implications of municipal finance policies in urban planning.

AI & Computational Modeling for Healthcare Applications

Jan 2022 – Jul 2022

Undergraduate Research Assistant (Prof. Kyung Eun Lee), Inha University, South Korea

- Achieved 98.6% accuracy in classifying five arrhythmia types from ECG signals, surpassing the state-of-the-art benchmark (97.3%), using a CNN model built in TensorFlow.
- Quantified how anastomosis angle and graft porosity affect vascular bypass patency using computational fluid dynamics, showing that larger angles reduce low wall shear stress regions and may lower restenosis risk.

Publications and Posters

- Kunwoo Lee**^{*}, Soyong Shin^{*}, Keelan Enseki, Adam Popchak, Jessica Hodgins, Eni Halilaj, *Domain Knowledge Improves AI-Based Evaluation of Physical Therapy Performance*. (In preparation) **[Manuscript]** (^{*} **Equal contribution**)
- Kunwoo Lee**^{*}, Chaeun Lee^{*}, Zhixiong Li, Vu Phan, Evy Maeinders, Eni Halilaj, *Does Markerless Motion Tracking Hold Up Under Load? Evaluation of Multi-View Methods for Lower-Extremity Joint-Moment Estimation*. Journal of Biomechanics, (In preparation) **[Manuscript]** (^{*} **Equal contribution**)
- Kunwoo Lee**, Dhivya Sreedhar, Pushkar Saraf, Chaeun Lee, Kateryna Shapovalenko, *Typing Reinvented: Towards Hands-Free Input via sEMG*. arXiv preprint, 2025 **[Manuscript]**
- Zhixiong Li, Chaeun Lee, **Kunwoo Lee**, Evy Meinders, Andrew L Sprague, James J Irrgang, Volker Musahl, Eni Halilaj, *Videos Capture Functional Strength Deficits Following ACL Reconstruction*. American Society of Biomechanics, 2025 **[Conference Poster]**
- Youngjo Choi, Lane Hartwig, **Kunwoo Lee**, Nam Gyu Lee, Juhyun Jung, Andres Iniguez, Zaid Iqbal, Darshan Sasidharan Nair, Jean-Yves Thompson, *Understanding the Varied Economic Impact of Tax Increment Financing (TIF)*. Illinois Institute of Technology Innovation Day, 2024 **[Poster Presentation]**
- Kunwoo Lee**, Scott T.M. Dawson, *Estimating Pitching Airfoil Kinematics Using Convolutional Neural Networks*. CAURS, 2025 **[Poster Presentation]**
- Kunwoo Lee**, SeokJu Lee, Woodam Kim, Kyung Eun Lee, *The Effects of Anastomosis Angle on Blood Flow in Bypassed Blood Vessel*. Biomedical Engineering Society for Circulation, 2022 **[Conference Poster]**

Graduate Coursework

Intro to Deep Learning (11-785)	Learning for 3D Vision (16-825)
Biomechanics of Human Movement (24-663)	Computer Vision for Engineers (24-678)
Robot Dynamics and Analysis (24-760)	

Skills

Programming:	Python, C++	ML/AI Frameworks:	PyTorch, TensorFlow
Cloud/HPC:	Slurm, AWS	Biomechanics Tools:	OpenSim, Theia3D, Visual3D, OpenCap
Visualization:	AitViewer, PyTorch3D		

Professional & Leadership Experience

Unit Supplier & Squad Leader

Jan 2018 – Sep 2019

ROK Army 2nd Corps, Chuncheon, South Korea

- Managed food supply logistics for unit operations, including ordering, inspection, transport, and distribution.
- Selected as a Designated Counseling Soldier to support personnel resilience under high-stress conditions, strengthening morale and cohesion through structured team-building practices.

Awards & Honors

Summa Cum Laude	Illinois Institute of Technology	2024
Dean’s List	Illinois Institute of Technology	2022–2023
TOEFL Exchange Student Scholarship	ETS Korea	2022
Honor Student	Inha University	2020–2021
Academic Excellence Scholarship	Inha University	2020–2021
Soldier of Valor Award	Republic of Korea Army	2019